F24PW417-VS
VARIABLE SPEED DRIVE

RD WH BU
11,000V
FCF 160A
QAC 160A 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10002

OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3
CABLE SCREEN EARTHED
F24-PW-417
HIGH PRESSURE PROCESS WATER PUMP 1
3,300V
800 kW

F24-PW-417
MOTOR HEATER TERMINAL BOX
X2 54 53
F24PW417HT-L
F24PW417HT-N
0.3kW
ANTI-CONDENSATION HEATER

MOTOR SPACE HEATER CIRCUIT SHEET 6
X81 (PART OF) RTD TEMPERATURE MONITOR RELAY SHEET 6

F24-PW-417
MOTOR RTD TERMINAL BOX

PT100 RTD INPUT CH1	+ 13	F24PW417A-RD	MOTOR WINDING RTD (RD PHASE) (TD1)	
	COM 14	F24PW417A-WH		
	- 15	F24PW417A-BK		
PT100 RTD INPUT CH2	+ 16	F24PW417B-RD	MOTOR WINDING RTD (WH PHASE) (TD3)	
	COM 17	F24PW417B-WH		
	- 18	F24PW417B-BK		
PT100 RTD INPUT CH3	+ 19	F24PW417C-RD	MOTOR WINDING RTD (BU PHASE) (TD5)	
	COM 20	F24PW417C-WH		
	- 21	F24PW417C-BK		
PT100 RTD INPUT CH4	+ 22	F24PW417D-RD	MOTOR BEARING RTD (DRIVE END) (BD10)	
	COM 23	F24PW417D-WH		
	- 24	F24PW417D-BK		
PT100 RTD INPUT CH5	+ 25	F24PW417E-RD	MOTOR BEARING RTD (NON DRIVE END) (BD20)	
	COM 26	F24PW417E-WH		
	- 27	F24PW417E-BK		
PT100 RTD INPUT CH6	+ 28			
	COM 29			
	- 30			
PT100 RTD INPUT CH7	+ 31			
	COM 32			
	- 33			
PT100 RTD INPUT CH8	+ 34			
	COM 35			
	- 36			

X81 pins 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 — 1139 — X2 67
X3 17 — 7703 — X2 68 — K41 — K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
X2 4 — 1179 — X2 63 — K41 — K41 — K41
X2 25 — 1199 — X2 64
MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
X2 3 — 1219 — X2 65
X2 24 — 1239 — X2 66 — K41 — K41 — K41
MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 — 1259 — X2 70
X2 44 — 1279 — X2 69
A30 (PART OF) X2 15 — TRIP — X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R10 — 25A
W10
B10
EARTH BAR
415VAC
F16DB101-R10 F16DB101-W10 F16DB101-B10 F16DB101-BK10
X4 1 2 3 4 PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
BU BK
B6 — 10A
EARTH BAR
240VAC
F16DB201-B6 F16DB201-BK6
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-002
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45
A2 (PART OF) DI-2B 9
X2 13 14
24VDC CONTROL SHEET 6

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F24PW417VS-1409
F24PW417VS-1419
F24PW417VS-1429
F24PW417VS-1439

F24PW417-LCS
LOCAL CONTROL STATION
X1 1 2 3 4 LOS
X1 5 6 E START
F24PW417VS-1459
F24PW417VS-1469

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F24PW417-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- ⊗ 11kV SWITCHBOARD TERMINAL
- ⊗ VSD TERMINAL
- ◒ FIELD TERMINAL
- ◑ DCS OR I/O TERMINAL
- ◧ MARSHALLING TERMINAL
- (x) DEVICE TERMINAL

PANEL A16

2000-F16-SCM-EL-10131_Rev1

Rougher & scavenger flotation F16-SW-011
F24-PW-417 hp process water pump 1 VSD feeder schematic diagram